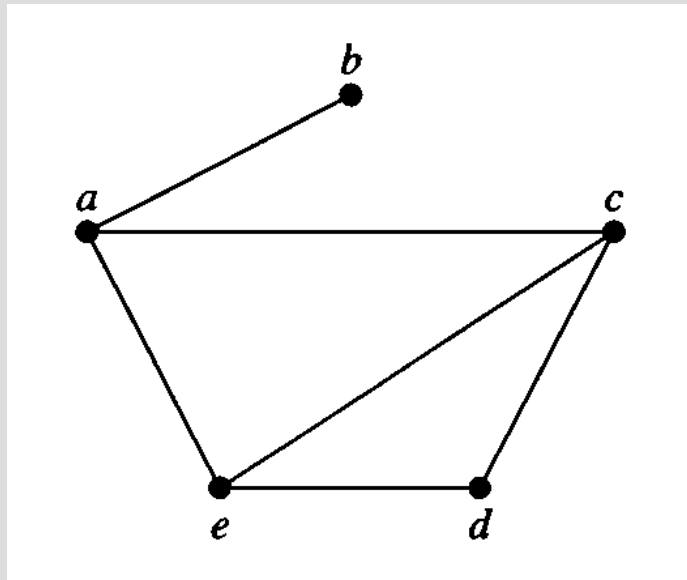


GRAPHS

CSI 219 – Discrete Mathematics

GRAPH REPRESENTATION

ADJACENCY LIST



Vertex	Adjacent vertices
--------	-------------------

<i>a</i>	<i>b, c, e</i>
----------	----------------

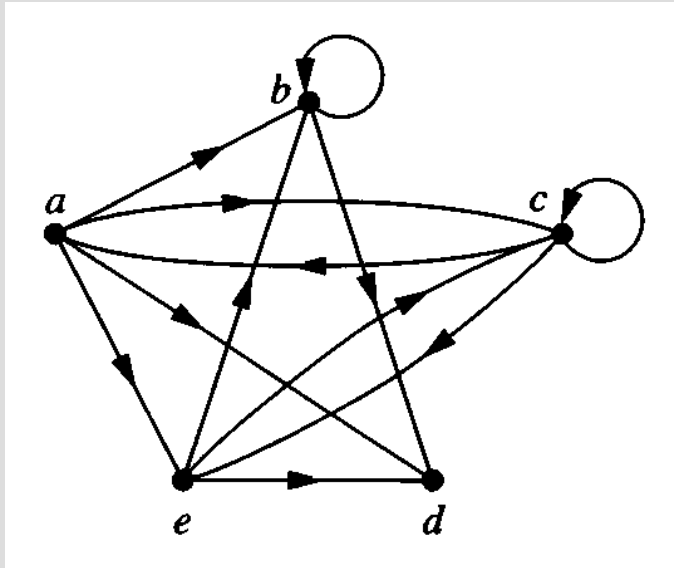
<i>b</i>	<i>a</i>
----------	----------

<i>c</i>	<i>a, d, e</i>
----------	----------------

<i>d</i>	<i>c, e</i>
----------	-------------

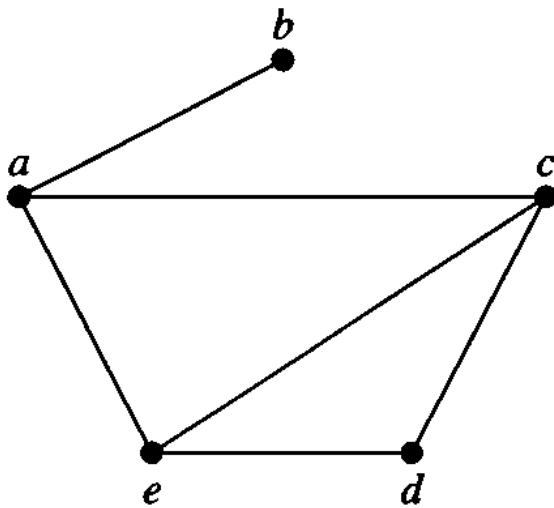
<i>e</i>	<i>a, c, d</i>
----------	----------------

ADJACENCY LIST



Initial	Terminal vertices
<i>a</i>	<i>b, c, d, e</i>
<i>b</i>	<i>b, d</i>
<i>c</i>	<i>a, c, e</i>
<i>d</i>	
<i>e</i>	<i>b, c, d</i>

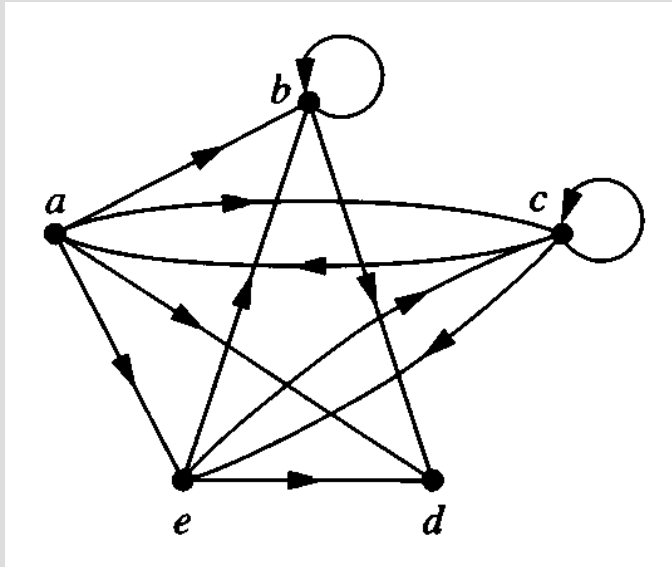
ADJACENCY MATRIX



	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>
<i>a</i>	0	1	1	0	1
<i>b</i>	1	0	0	0	0
<i>c</i>	1	0	0	1	1
<i>d</i>	0	0	1	0	1
<i>e</i>	1	0	1	1	0

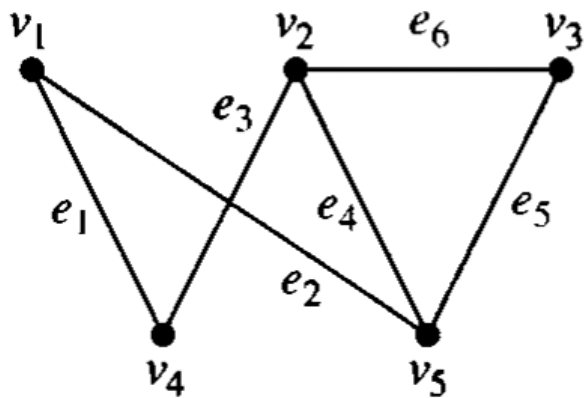
What would happen if more than one edge between two vertices?

ADJACENCY MATRIX



	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>
<i>a</i>	0	1	1	1	1
<i>b</i>	0	1	0	1	0
<i>c</i>	1	0	1	0	1
<i>d</i>	0	0	0	0	0
<i>e</i>	0	1	1	1	0

INCIDENCE MATRIX



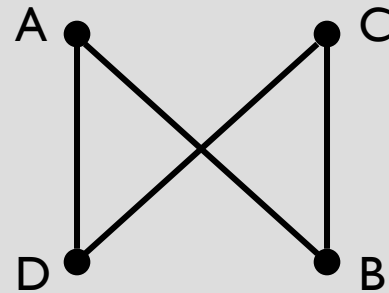
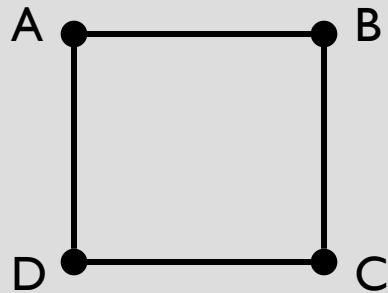
$$\begin{matrix} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix}.$$

PRACTICE PROBLEM

- Chapter-9.3

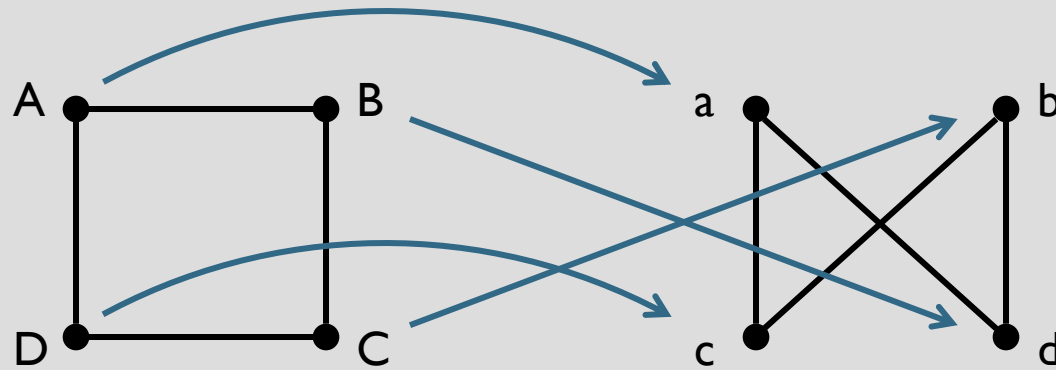
4, 8, 9(e), 11, 17

GRAPH ISOMORPHISM



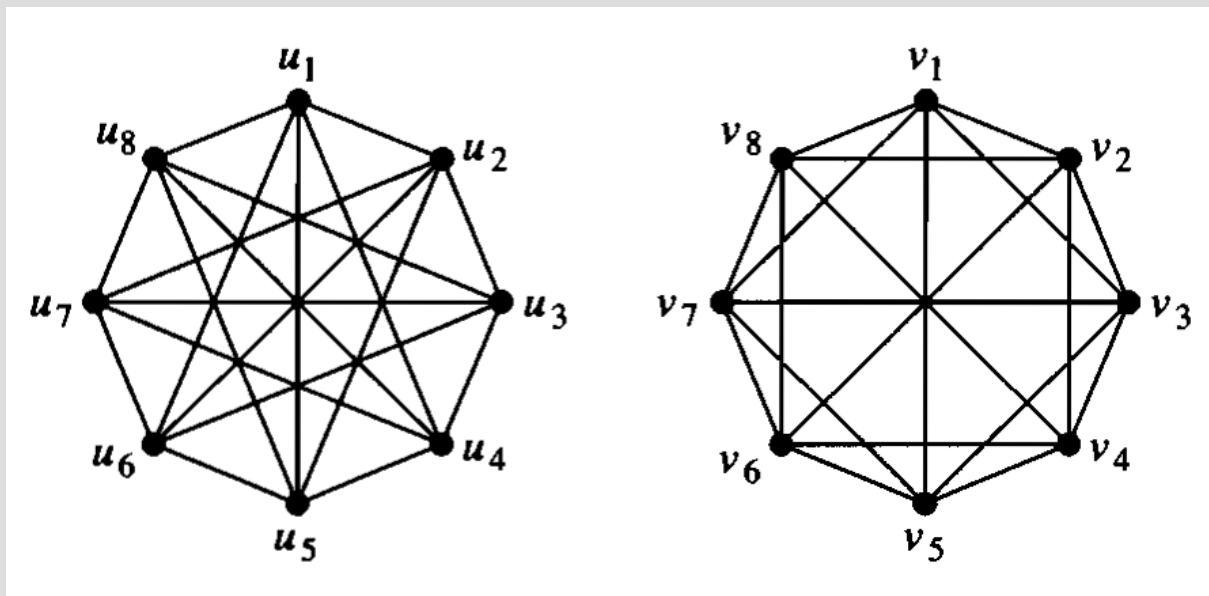
- We can draw the same graph differently
- If we can determine that two graphs are actually two different representations of the same graph, then they are called isomorphic

GRAPH ISOMORPHISM



- Theoretically, two graphs are isomorphic if there is a one-to-one and onto function f from V_1 to V_2 (the sets of vertices) such that a and b are adjacent in G_1 if and only if $f(a)$ and $f(b)$ are adjacent in G_2 , for all $a, b \in V_1$

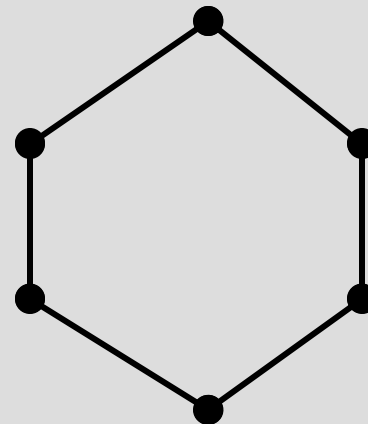
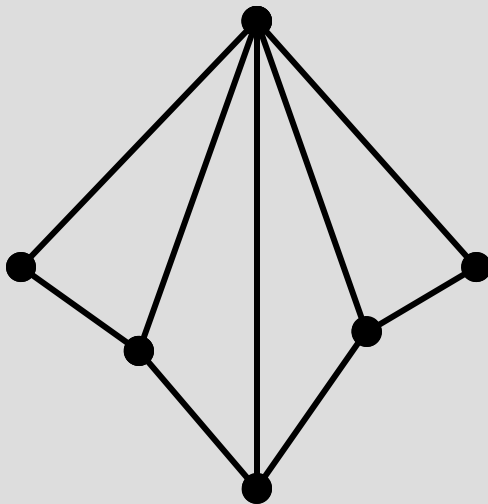
ARE THESE GRAPHS ISOMORPHIC?



DETERMINING ISOMORPHISM

- Determining isomorphism is a hard problem
 - $n!$ possible one-to-one correspondences between the vertex sets of two simple graphs with n vertices
- But sometimes, it is not hard to show that two graphs are NOT isomorphic

ARE THESE GRAPHS ISOMORPHIC?



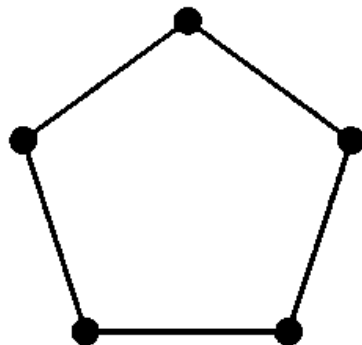
DISPROVING ISOMORPHISM: GRAPH INVARIANTS

- We can show that two graphs are not isomorphic if we can find a property that only one of the two graphs has, but the property is preserved by isomorphism
 - Such a property is called a [graph invariant](#)
- For example, two isomorphic graphs must have the same number of vertices, and the same number of edges as well

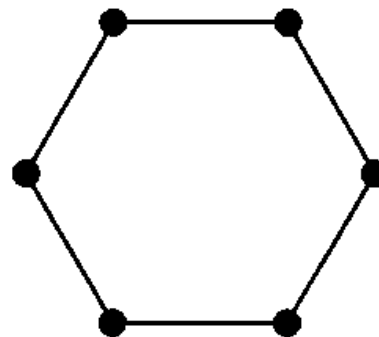
GRAPH INVARIANTS

- Number of vertices
- Number of edges
- Degrees of each vertex

ARE THESE GRAPHS ISOMORPHIC?

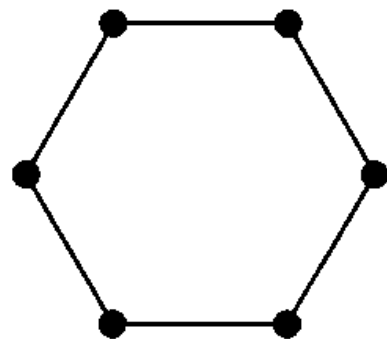


C_5

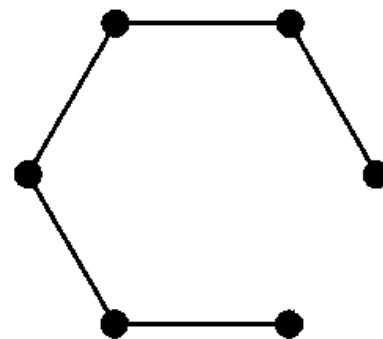


C_6

ARE THESE GRAPHS ISOMORPHIC?

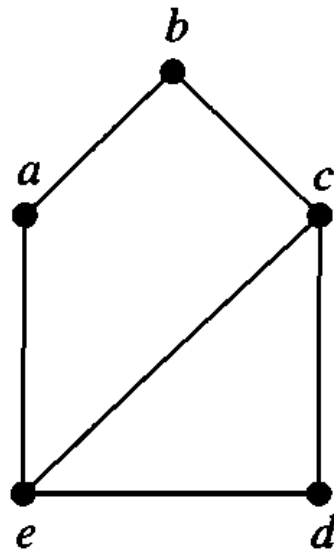


C_6

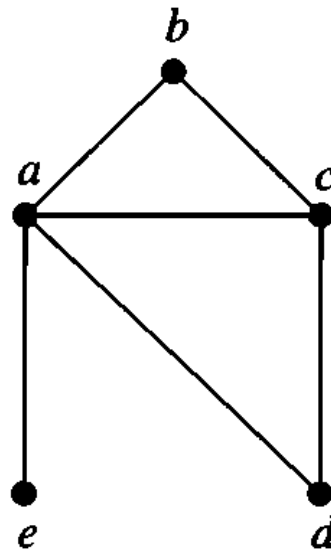


C_6

ARE THESE GRAPHS ISOMORPHIC?



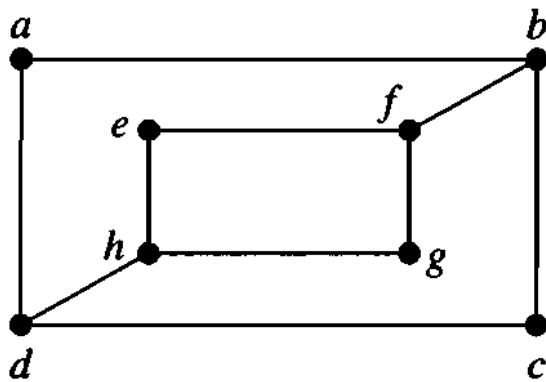
G



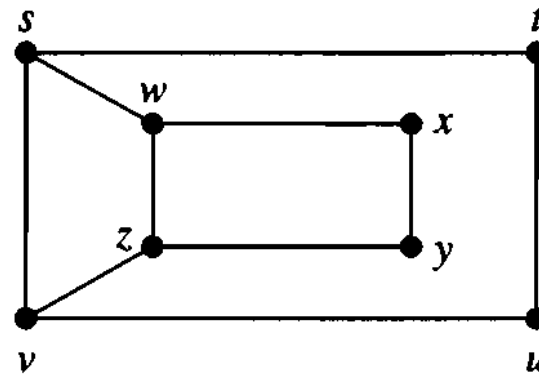
H

ARE THESE GRAPHS ISOMORPHIC?

2, 3, 2, 3, 2, 3, 2, 3



G



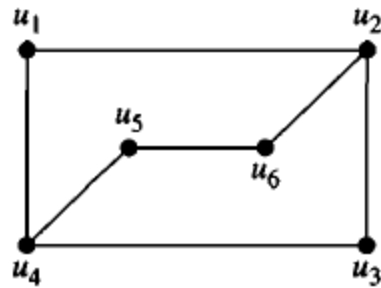
H

3, 2, 2, 3, 3, 2, 2, 3

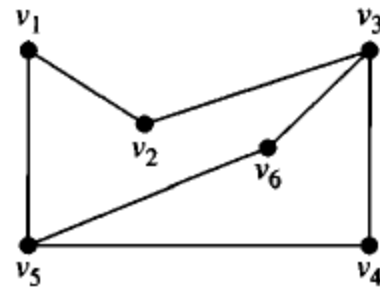
GRAPH INVARIANTS

- Number of vertices
- Number of edges
- Degrees of each vertex
- Degrees of neighbor vertices

EXAMPLE



G



H

SOLUTION

$$f(u_1) = v_6, f(u_2) = v_3, f(u_3) = v_4, f(u_4) = v_5, f(u_5) = v_1, f(u_6) = v_2.$$

EXAMPLE

$$\mathbf{A}_G = \begin{matrix} & \begin{matrix} u_1 & u_2 & u_3 & u_4 & u_5 & u_6 \end{matrix} \\ \begin{matrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix},$$

$$\mathbf{A}_H = \begin{matrix} & \begin{matrix} v_6 & v_3 & v_4 & v_5 & v_1 & v_2 \end{matrix} \\ \begin{matrix} v_6 \\ v_3 \\ v_4 \\ v_5 \\ v_1 \\ v_2 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}.$$